

Alju Dress: Pets as a metaphor for wearable display designs

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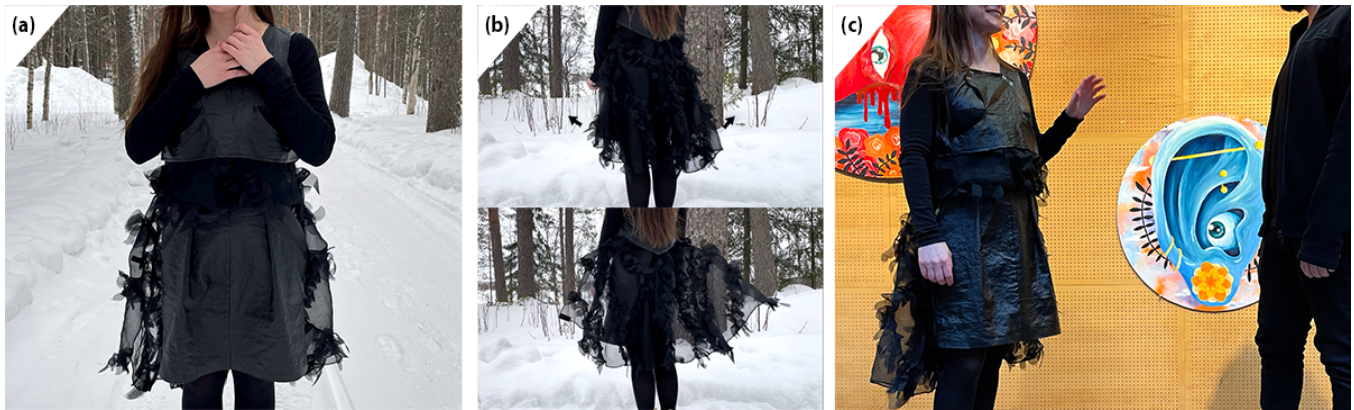


Figure 1: (a) The Alju Dress in the forest; (b) two states of the Alju's wings; (c) the Alju dress in a cocktail setting.

ABSTRACT

Alju is inspired by the pet-owner relationship and mimics exotic birds living on our bodies. It is a dress that interacts with sudden changes in its surroundings by reacting to the sounds around it. It indicates moderate and high noise levels by imitating bird wings moving on the garment in different velocities. Just like a pet, it acts ambiguously and requires us to pay attention to its intentions and try to interpret why it acts the way that it does. In this regard, our work aims to present pets as a metaphor for designing wearable displays and transferring animal-like behaviors to a garment to replicate emotional communication between pets and their owners, easing the way the wearer perceives the challenges of wearable displays automatically modifying her appearance.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; **Robotics**; • **Networks** → *Network reliability*.

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KEYWORDS

Pet, Wearable Displays, Self-expression, Fashion, Social Wearables, Ambient Displays

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1 INTRODUCTION

Bringing technology to our bodies opened up various possibilities for garments that are aware of the wearer and the environment around them, as well as reacting to these conditions. They are capable of transforming their states with movement, light, sound, vibration and so on. However, the design of wearable displays should consider social challenges for the wearers as these devices share the same space with traditional clothes and accessories on our bodies. The automatic changes in the wearable, when observable from outside, might create conflicts regarding the wearer's perception of her image or the appropriateness to the context they were worn [7–9].

In our work, we propose an alternative design perspective to wearable displays for conveying information. We ask the question if we can consider *pets* as a metaphor for designing wearable displays. These pets may react without our consent, yet just like pets, we can pay attention to their intentions and try to understand why they act the way that they do so. We argue that designing pet-like

interactions on wearables can ease up the social challenges of the wearer's appearance automatically altering. To illustrate this, we created the Alju (Figure 1), a dress that uses pet birds on the body as a metaphor for displaying information. It creates movements on the dress that mimics the birds flocking wings and interacting with the noise level of environment. It creates an emotional channel between the garment and the wearer where the wearer becomes the owner of a living dress. In this respect, our work contributes a starting point for designing a platform that considers pet-owner relationship as a metaphor for ambient information-giving on the body.

2 RELATED WORK

Having animals as companions in households is exceptionally growing in popularity. In the USA, there are about 60% of the population keeps pets [2]. Keeping these animals is argued to fulfill several purposes: The owners associate emotions to the pets' behaviors [23, 26]. The research suggests this communication between pets and their keepers has health benefits such as providing companionship and social support [14, 22]. It is also evident that the owners see pets as a way of making a statement or extending the self [17]. Here, an extreme example can be named by mentioning Salvador Dali walking his anteater in Paris as being a symbolic statement about the surrealism movement [1]. Tamagotchi might be the first example of an interactive device featured as a virtual pet [3]. Also, pet-like features inspired interactive artifacts, more specifically robots [19]. Coghlan et al. [6] argue that robot pets might support the elderly as robots can be custom-designed companions. Furthermore, by evaluating a furry robot, Bucci et al. [4] suggest ambiguity of kinetic movements in a pet-like robot might evoke emotions in observers.

In the wearable field, Behnaz Farahi [10] demonstrated the kinetic movements of animal-like outputs (resembling fur) on a shoe as an emotive display. Furthermore, in a survey examining the perception toward on-body robot companions, Jiang et al. [16] reported that animal-like robots might be useful as a tool or an assistant to their users. Moreover, Kao et al. [18] presented their kinetic robots moving on the body to designers and potential wearers, noting that the movement of the pieces are perceived as animals and can be used both for aesthetic (i.e. shape and pattern changing clothes) and functional purposes (i.e. conveying notification from mobile phones). Spider Dress[25] and Butterfly Dress [20] might be given as expressive uses of the wearable animal-like displays on the body. Whereas the Spider dress moves the spider arms on the body for preserving personal space, on the Butterfly dress, the butterflies departs from the body, for example when a bride says yes at a wedding, complementing the emotional state of the moment. In our work, we demonstrate "pets living on our body" as a metaphor for conveying ambient information.

3 CONCEPT

3.1 Design Process & Implementation

The Alju Dress (Figure 1) aims to mimic the owner-pet relationship by transferring pet-like interactions to garments. Our inspiration for the concept came from the wild birds (magpies, crows, seagulls and hawks) that were domesticated by many households in Northern

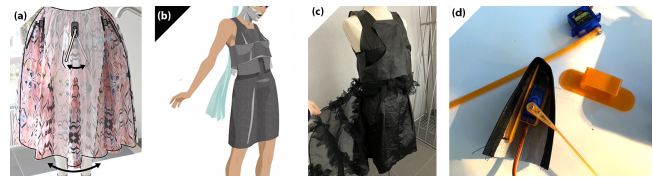


Figure 2: Construction and design process of the Alju dress: (a) First prototype of Alju concept as a skirt; (b) first sketches of the Alju dress; (c) the Alju dress before the electronics placed; (d) servo-motors and 3D casings used in the Alju dress.

Finland back in the '70s. Often these animals became mascot-like companions to their owners. The name for them in the Northern Finnish dialect is "alju". It means "tame". Birds on the body have been a focus of attention of fashion designers and artists before. One of the first example of this concept was apparent in the carnivals at Nice during the nineteenth century, i.e. a bird cage costume with birdcage paintings symbolizing Bohemianist live styles of the wearers [5]. A more recent interpretation took form in Kasey McMahon's wearable sculpture called Birdcase Dress [21]. This dress has metal framings shaped like a cage-like skirt as a functional cage. These works uses birds on the body as unexpected figures residing on the body. Our work takes the oddness of wearing birds further by focusing on their movement on the body for conveying information.

Our design process started with the idea of birds living on a body that is sensitive to the environment. We chose noise as input since noise pollution is an important issue, causing annoyance and distraction or even chronic stress and sleep deprivation in long-time exposures [24]. We imagined the birds can react to the level of ambient noise by flicking their wings to inform both the wearer and the observers. The first prototype of the Alju was designed as a skirt that reacts to noises by mimicking birds' movements under the skirt (Figure 2-a).

In the second iteration, we took the visual features of the birds into consideration with wing-like structures on its sides. In this version, the user creates a surprising and mystical atmosphere around them with this interactive "smart / business casual" suit leaning in a futuristic Gothic style. The shape of the garment began to emerge largely around translucent material (feathery netted fabric) used for the pockets and wings. This fabric complements the animal's visual features with its black, shiny feather cover like on a crow or a raven. The movement of the wing-like structures also reflects the functionality. They are fixed on the backside of the waist. The loose parts on the sides are connected to 3D printed sticks, attached to two servo-motors placed on the sides of the dress by using 3D printed casings (Figure 2-d). This structure allowed to move the wings up and down by activating the motors (Figure 1). The movement reflects the reaction of the bird to environmental sound levels: They stay still when it is quiet, whereas they start flipping as a reaction to environmental sound. For instance, the dress flips its wings slowly in a cocktail setting when people talking (Figure 1-c) and it flips the wings quickly when there is a sudden high noise like people clapping.

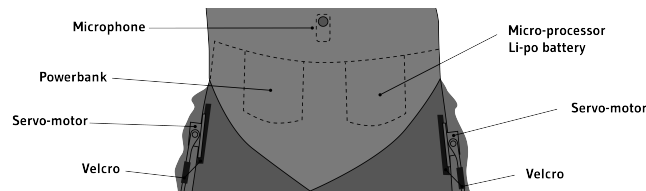


Figure 3: An illustration showing the components and their placements on the waist part of the dress.

The interactive functionality of the garments is achieved with electronic components including a microprocessor (Arduino MKR Wifi 1010), two servo motors (SG90), a USB powerbank (5V) to power servo-motors, a Li-po battery (3.7V) to power the rest of the system and a SparkFun Electret Microphone Breakout module to detect environmental sound level. The microprocessor and the batteries are placed in small pockets on the backside of the garment, whereas the microphone was fixed on the upper backside of the garment. The motor casings are sewn on pieces of the garment's base fabric and they were fixed on sides of the garment by using Velcro. Figure 3 shows the component placements.

The Arduino MKR Wifi 1010 was programmed to sense two sound levels (moderate and high) by using the microphone: When the sound level is below moderate, the servo motors do nothing. When it is in moderate and high ranges, all motors rotate causing wings to move up and down. In moderate sound levels, motors slowly rotate, creating a calm, wavy movement on the drapes. As for sudden and loud noises, the motor movements switch to a higher speed leading to sharper and quick movements. The moderate sound levels were defined by experimenting with people speaking nearby whereas an object falling down to the ground or somebody clapping was used to determine high noise. While the current state of the garment displays only sound levels, the usage of a microprocessor with wireless communication capabilities was intended to expand our approach with other information input possibilities (i.e. phone notifications).

4 DISCUSSION & CONCLUSION

Finding suitable metaphors to justify the visual alterations on the wearable displays is a challenging task [8, 11–13, 15]. Without a careful design considering the wearer's image or the appropriateness to the context, the wearable displays might be rejected due to lack of appropriateness [7, 9]. We argue a pet metaphor and transferring animal-like behaviors to a garment can replicate emotional communication between pets and owners, easing the way the wearer perceives her appearance automatically modifying. For this, we created Alju, a dress that reacts to the noise level of the environment by mimicking birds' movements. The dress blends the bird metaphor in the holistic design of the garment and warns us, in its own way, by flicking its wing-like structures. In addition to informing wearer and others around, we expect the ambiguous and odd movements of the dress can also open up new ways of expressing the self in a social context. In this respect, our work contributes a design approach as a starting point to examine the pet-owner relationship as a metaphor for designing wearable ambient displays.

While we demonstrated an output modality using a pet metaphor, it neglected the interactions that could be transferred from the owner's end to the pet. Future work might exemplify other interactions as petting, feeding, etc. for controlling and adjusting the wearable display. Also, in its current version, the input is limited to environmental sound. In the future, we plan to experiment with different types of information like phone notifications and evaluate our approach in in-the-wild studies examining the social and individual experiences that our prototype could yield.

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